

Process Verification

Lab

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Chapter 1

Design Overview

1.1 Controller

A dual-master memory controller. Master 0 (CPU) has fixed priority over Master 1 (DMA). Synchronous writes on clock, and combinational reads.

1.1.1 Parameters

There are two parameters to this dut:

- `ADDR_WIDTH` : address size.
- `DATA_WIDTH` : data width.

1.1.2 Interface

1.1.3 Functional Description

- **Reads:** 0 clock cycles (combinational —immediate once address & grant are stable).
- **Writes:** 1 clock cycle (synchronous —update happens on next rising edge when grant & we are true).

Signal	Direction	Width	Description
clk	Input	1	Clock signal
rst_n	Input	1	Active-low synchronous reset
addr0	Input	ADDR_WIDTH	Master 0 address
wdata0	Input	DATA_WIDTH	Master 0 data
rdata0	Output	DATA_WIDTH	Master 0 data
we0	Input	1	Master 0 write enable
req0	Input	1	Master 0 read request
gnt0	Output	1	Master 0 grant
addr1	Input	ADDR_WIDTH	Master 1 address
wdata1	Input	DATA_WIDTH	Master 1 data
rdata1	Output	DATA_WIDTH	Master 1 data
we1	Input	1	Master 1 write enable
req1	Input	1	Master 1 read request
gnt1	Output	1	Master 1 grant

Table 1.1: Controller signals

Condition	Behavior
Only one master valid	That master is serviced immediately
Both masters valid	Master 0 is serviced (priority). Master 1 is not granted until Master 0 de-asserts request
No master valid	Controller idle

Table 1.2: Controller Behavior

Chapter 2

Diagram

Single master transaction:

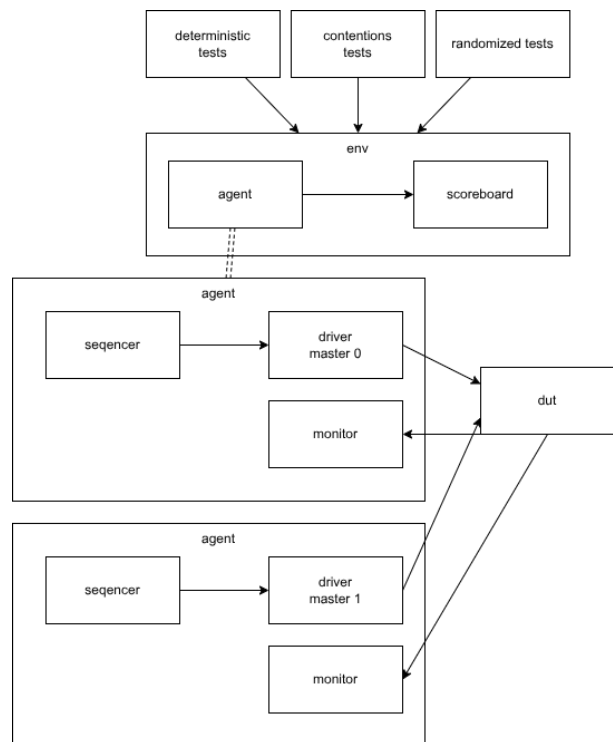
```
class ctrl_master_transaction extends uvm_sequence_item;
    'uvm_object_utils(ctrl_master_transaction)

    rand bit use_m;
    rand bit is_write;
    rand logic [ADDR_WIDTH-1:0] addr;
    rand logic [DATA_WIDTH-1:0] data;

    function new(string name = "ctrl_master_transaction");
        super.new(name);
    endfunction

    function string convert2string();
        return $sformatf(
            "M: [%0s] Addr=%0h Data=%0h",
            (use_m ? (is_write ? "WR" : "RD") : "IDLE"), addr, data,
        );
    endfunction

endclass
```



Appendix A

Appendix Chapter

A.1 Use of tools

- Visual Studio Code: systemverilog IDE.
- QuestaSim: CLI tester.
- GTKwave: wave visualizer.

A.2 Use of time

A total of 1h distributed as follows.

- Understanding the problem: 1h.
- Report: 1h.

A.3 Use of AI

- Gemini (Pro): sentence enhancing, bug fixer